Sensation and Perception

How the senses turn the world into signals, why small changes go unnoticed, and why the same product tastes better with a higher price on it.

Isagi · November 2026 · Spine: Ciccarelli & White, *Psychology* (4th ed.), Ch. 3 — rewritten, updated and applied

Marketing is applied perception. A customer never meets your product directly; they meet what their senses report and what their brain makes of it. A pack shrinks by 8% and nobody notices; a price rises by 8% and everyone does. A paint shade looks different under the store's tube light than on the living-room wall. This part explains why, and gives you two numbers-based tools — the difference threshold and Weber's law — that pricing and packaging decisions already use, whether or not anyone names them.

1 Sensation versus perception

Sensation is the raw input: light hitting the eye, pressure on the skin, molecules on the tongue. Each sense organ converts its kind of energy into nerve signals — a step called **transduction**. **Perception** is what the brain does with those signals: organising them, interpreting them, and giving them meaning. Sensation tells you there are red and white shapes on a shelf. Perception tells you it's a can of Coke, and that you want one.

The distinction matters because most marketing battles are won or lost at the perception stage. Two products can deliver identical sensations and be perceived very differently.

2 Thresholds: what we notice

The **absolute threshold** is the smallest amount of a stimulus a person can detect half the time — the faintest sound, the dimmest light. The more commercially useful number is the **difference threshold**, or **just noticeable difference (JND)**: the smallest *change* a person can detect half the time.

In the 1830s Ernst Weber found that the JND is not a fixed amount but a fixed *proportion* of the original. This is **Weber's law**:

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ΔI = k × I
ΔI = the just noticeable change
I  = the starting intensity
k  = Weber fraction (constant for a given sense)
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If you're holding a 1 kg bag, you might need about 20 g more to notice the difference; holding 10 kg, you'd need about 200 g. The fraction differs by sense — we are much more sensitive to some changes than others — but the proportional logic holds over a wide middle range.

WORKED EXAMPLE 1 · SHRINKFLATION AND WEBER'S LAW

A biscuit brand sells a 100 g pack. Input costs rise. Suppose consumer tests show that people notice a change in pack weight about half the time once it reaches 10% (k = 0.10; illustrative).

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JND = k × I = 0.10 × 100 g = 10 g

Cut to 95 g → change 5 g  < 10 g → mostly unnoticed
Cut to 88 g → change 12 g > 10 g → mostly noticed

Price: ₹10 → ₹11 is a 10% change, and a whole
rupee on a round number is very easy to notice.
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This is why brands shrink packs in small steps instead of raising round-number prices. The same law works in reverse for improvements: a "new, improved" change must clear the JND, or customers won't perceive any difference and the investment is wasted. Weber's law tells you how big a visible change has to be.

Shrinkflation is legal in India if the label states the true net quantity, as the Legal Metrology rules require. It is also widely reported in the press — customers notice eventually, especially when someone points it out.

Sensory adaptation is the senses' tendency to stop responding to an unchanging stimulus. You stop smelling your own perfume within minutes, and stop hearing the ceiling fan. Related is **habituation**, where the brain learns to ignore a repeated, unimportant stimulus — the reason the same banner ad in the same spot stops being seen. Change is what grabs attention.

3 Vision

Light enters through the **cornea** and **pupil**; the **lens** focuses it onto the **retina** at the back of the eye. The retina has two kinds of receptor:

- **Rods** — about 120 million, very sensitive to dim light, no colour, mostly in the edges of vision.

- **Cones** — about 6 million, need brighter light, see colour and fine detail, concentrated in the centre (the **fovea**).

Where the optic nerve leaves the eye there are no receptors at all — the **blind spot**. You never notice it because the brain fills in the gap, an early hint of how much perception is construction.

Colour is explained by two theories that are both right, at different stages. The **trichromatic theory**: three types of cone respond most to short (blue), medium (green) and long (red) wavelengths, and every colour is a mix of their signals. The **opponent-process theory**: further along, cells work in opposing pairs (red–green, blue–yellow, black–white). Stare at a red patch, then look at a white wall, and you see a green **afterimage** as the tired red signal lets green dominate.

Red–green colour blindness is common in men — roughly 1 in 12 in many populations. Practical point: a sales dashboard that shows good and bad territories only in red and green is unreadable to a meaningful share of your managers. Add labels, shapes or a blue–orange palette.

4 Hearing

Sound is vibration in the air. Its **frequency** is heard as pitch, its **amplitude** as loudness (measured in decibels, dB), and its mix of frequencies as **timbre** — why a flute and a violin playing the same note sound different. The decibel scale is logarithmic: every 10 dB is ten times the sound intensity. Long exposure above about 85 dB — heavy traffic, a loud factory floor — can permanently damage hearing, which is a real occupational issue in manufacturing plants.

5 Taste, smell, touch and pain

Taste receptors detect five basic tastes: sweet, sour, salty, bitter and **umami** (savoury, the taste of glutamate). Most of what we call "flavour", though, is smell. Hold your nose and an apple and a raw onion are surprisingly hard to tell apart.

Smell has an unusual route: it is the one sense that doesn't pass through the thalamus first, and it connects closely to the limbic system's emotion and memory areas. That is the logic behind scent marketing — hotel lobbies, car showrooms, the bakery smell near a supermarket entrance. The effects in studies are real but usually modest, and they depend on the scent fitting the product.

The skin senses pressure, temperature and pain. The **gate-control theory** of pain says signals pass through a "gate" in the spinal cord that other inputs can partly close: rubbing a bumped elbow really does reduce the pain. The brain can also close the gate from above — attention and expectation change how much something hurts.

6 Perception: how the brain organises

The Gestalt psychologists (Part 0) showed that we organise sensations into wholes using predictable rules:

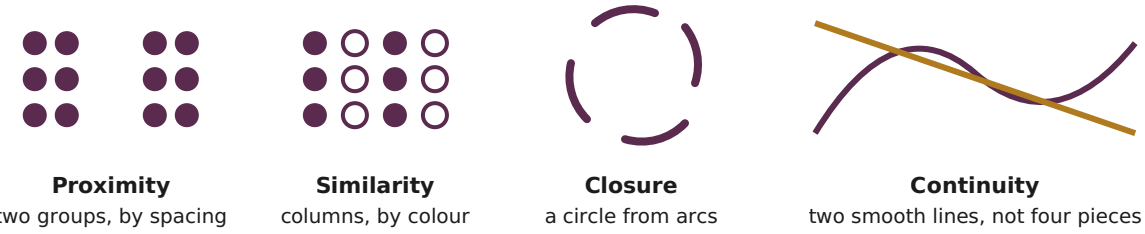


Figure 1. Four Gestalt principles. We see groups, columns, a circle and two crossing lines, though none is "really" there.

- **Figure–ground** — we separate an object from its background. Packaging that doesn't stand out from the shelf behind it fails at this first step.
- **Proximity** — things close together are seen as a group. Products shelved together are read as a range.
- **Similarity** — things that look alike are grouped. A consistent pack colour across a range is "brand blocking".
- **Closure** — we complete incomplete shapes. Several well-known logos rely on it.
- **Continuity** — we follow smooth lines and patterns.

The brain also holds things constant as the input changes. **Size constancy**: a person walking away doesn't seem to shrink. **Colour constancy**: a white shirt looks white under yellow bulbs and blue daylight, though the light reaching your eye is very different. Constancy has limits, which matters in a paint store — under tube lights a warm shade can read cooler than on the customer's wall, which is why shade cards and sample pots exist.

Depth comes from both eyes together (each sees a slightly different view) and from single-eye cues artists use: nearer objects overlap farther ones, parallel lines converge in the distance, distant things look hazier and smaller. **Illusions** such as the Müller-Lyer lines (two equal lines that look different lengths because of the arrow-fins on their ends) show these normally helpful rules misfiring.

7 Expectation shapes experience

Bottom-up processing builds perception from the raw features upward. **Top-down processing** uses what you already know and expect to interpret them. Both run all the time, and top-down influence is far stronger than most people assume. A **perceptual set** is a readiness to perceive something in a particular way — and it changes not only what we judge, but what we actually experience.

The clearest commercial demonstration is a 2008 study by Hilke Plassmann and colleagues. Participants tasted wines in a brain scanner and were told the price. The same wine, labelled at $90 instead of $10, was rated more pleasant — and activity rose in a brain area linked to experienced pleasantness. The price did not just change the opinion; it changed the experience. Blind taste tests across many categories show the same pattern: labels and brands shift ratings even when the product is identical.

For a manager the implication cuts two ways. Presentation, price and brand are part of the product, not decoration around it. And internal product tests should be run blind, or your team's expectations will contaminate the result.

8 Held up? Subliminal advertising and other claims

In 1957 market researcher James Vicary claimed that flashing "Eat Popcorn" and "Drink Coca-Cola" on a cinema screen too briefly to see had raised popcorn sales by 58% and Coke by 18%. It caused a panic and calls for bans. In 1962 Vicary admitted the study's data were too small to mean anything, and it is widely regarded as fabricated.

Real research since then has found that a stimulus below awareness can have small, short-lived effects. In a 2006 Dutch study, a subliminal brand name nudged choice of a drink — but only among people who were already thirsty, and only briefly. There is no good evidence that subliminal messages can make people want something they didn't already want.

Claim	Verdict	Why
Hidden messages in ads control buying	**No**	The famous study was faked; real effects are tiny and need a matching need.
Price and brand change how a product tastes	**Yes**	Repeated in blind and branded tests; supported by brain-imaging work.
Small pack reductions go unnoticed	**Largely**	Weber's law is robust. Customers do eventually notice, often via media coverage.
Specific colours reliably cause specific emotions (red = urgency, blue = trust)	**Weak**	Colour effects exist but depend heavily on context and culture; universal "colour psychology" charts are mostly folklore.

WORKED EXAMPLE 2 · SAME RUPEE, DIFFERENT FEELING

A dealer gets ₹500 off on a ₹5,000 order and ₹500 off on a ₹50,000 order. Same rupees. Weber's law predicts very different perceived value, because perception tracks the proportion:

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₹500 / ₹5,000  = 10%  → clearly felt
₹500 / ₹50,000 =  1%  → barely registers
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To make a discount feel the same on the bigger order, you would need roughly 10% of it — ₹5,000. So large orders call for different levers: a free item, extended credit, or a gift that is valued in its own right, rather than a small cut to a large number.

Summary

- **Sensation** is detecting energy and converting it to nerve signals (**transduction**); **perception** is organising and interpreting those signals.
- The **JND** follows **Weber's law**: the noticeable change is a constant proportion of the starting level ($\Delta I = k \times I$). It explains small-step shrinkflation and why improvements must be large enough to be seen.
- **Sensory adaptation** and **habituation** make unchanging stimuli disappear; change grabs attention.
- Rods see dim light, cones see colour and detail. Colour works through three cone types and then opposing pairs. Around 1 in 12 men are red–green colour blind.
- Most flavour is smell; smell links directly to emotion and memory areas. Pain passes through a "gate" that other inputs and expectations can partly close.
- **Gestalt** principles — figure–ground, proximity, similarity, closure, continuity — govern how shelves and packs are read.
- **Top-down** expectations change actual experience: the same wine with a higher price tastes better. Run product tests blind.
- **Subliminal advertising** doesn't work as claimed; the 1957 study was faked.

Practice problems

P1. A 1 litre paint tin is to be reduced in volume. If k = 0.08 for perceived tin volume, what is the largest reduction that would mostly go unnoticed?

P2. R&D improves a detergent's whiteness by 3%. Consumer tests show the Weber fraction for whiteness is about 0.05. Will customers notice? What would you recommend?

P3. Name the Gestalt principle at work: (a) one brand's twelve products all in the same orange packs, (b) a promotion stand placed right next to the main brand bay, (c) a pack that blends into the shelf's colour behind it.

P4. Your team runs a taste test of a new snack against the leader, with both packs visible. The new snack loses. What's wrong with the test, and how would you fix it?

P5. An agency offers to embed hidden subliminal messages in your TV ad for a premium charge. Using the evidence, write your two-line reply.

P6. A ₹200 discount is offered on both a ₹2,000 product and a ₹20,000 product. Which will feel more generous? What discount on the expensive product would feel equivalent?

Glossary

Sensation / perception

Detecting physical energy / organising and interpreting it.

Transduction

Converting a stimulus into nerve signals.

Absolute threshold

The weakest stimulus detected half the time.

Just noticeable difference (JND)

The smallest change detected half the time.

Weber's law

The JND is a constant proportion of the starting stimulus: $\Delta I = k \times I$.

Sensory adaptation

Reduced response of the senses to an unchanging stimulus.

Habituation

The brain learning to ignore a repeated, unimportant stimulus.

Rods / cones

Retinal receptors for dim light / for colour and detail.

Trichromatic / opponent-process

Colour theories: three cone types / opposing colour pairs.

Gate-control theory

Pain signals pass a spinal "gate" that other input can partly close.

Gestalt principles

Rules by which the brain groups sensations into wholes.

Perceptual constancy

Seeing size, shape or colour as stable despite changing input.

Top-down processing

Interpreting input using expectations and knowledge.

Perceptual set

A readiness to perceive something in a particular way.

Subliminal
Below the threshold of conscious awareness.

Answer key

P1. JND = 0.08 × 1,000 ml = 80 ml. A reduction below about 80 ml would mostly go unnoticed (and the label must still state the true volume).

P2. No — 3% is below the 5% JND, so most customers won't perceive it. Either push the improvement past 5%, or don't build the campaign on the visible difference.

P3. (a) similarity; (b) proximity; (c) a figure–ground failure.

P4. Visible packs let brand expectations (top-down processing) change the taste experience. Run it blind: identical unmarked containers, randomised order, and ideally testers who don't know which is yours.

P5. For example: "The original 1957 subliminal study was admitted to be meaningless, and real effects since are tiny and short-lived. We'll spend the budget on what people can actually see."

P6. The ₹2,000 product: 10% versus 1%. An equivalent feel on the ₹20,000 product needs about 10%, roughly ₹2,000.

Sources: chapter structure follows Ciccarelli & White, *Psychology*, 4th ed. (Pearson), Ch. 3; text and figures are original. Wine and price: Plassmann et al., *PNAS*, 2008. Subliminal priming: Karremans, Stroebe & Claus, *Journal of Experimental Social Psychology*, 2006. Vicary's 1962 admission was reported in *Advertising Age*. Weber fractions in the worked examples are illustrative.